Exp No: 10	**Applying performance optimization techniques (indexing, subforms, query tuning) to a Zoho Creator application.**

AIM:

To apply the performance optimization techniques in a Zoho Creator application by using lookup / unique fields for fast searching (indexing), subforms for related data and tuned Deluge queries, and to compare the working of the application before and after the tuning.

ALGORITHM / PROCEDURE:

When the number of records in an application grows, the reports load slowly and the Deluge scripts take more time. The main techniques used to avoid this are: (a) **indexing** - Creator keeps the ID and the lookup fields ready for fast searching, so the criteria are written on these fields and the fields of unique values are marked *Unique*; (b) **subforms** - the child records (like order items) are kept inside the main record instead of a separate form linked again and again; and (c) **query tuning** - fetching only the needed records with criteria and range, avoiding fetch inside a loop and using the aggregate functions.

1. Create an application named *Order Management*. Add a form *Customers* (Customer_Name, Email - set as **Unique**) and a form *Orders* with the fields Customer (**Lookup** to Customers), Status, Order_Date and Amount.
2. Add about 500 to 1000 sample records in the *Orders* form (using the Import option) so that the difference in speed can be seen.
3. **Indexing:** use the Lookup field *Customer* and the Unique field *Email* in all the search criteria and report filters, instead of searching on the plain text fields.
4. **Subforms:** create one more form *Sales_Order* and add a **Subform** named Items (Product, Quantity, Price) in it, instead of creating a separate Items form and linking each row to the order. Check that the items are saved along with the main record in one submission.
5. Create a button / workflow with the **slow** script (Script 1) that fetches all the records and checks each record inside a loop. Run it and note the time taken from the execution log.
6. Now change the script to the **tuned** version (Script 2) that uses the criteria, the range and the aggregate functions. Run it again on the same data and note the time.
7. Open the report settings and reduce the load time of the report by showing only the needed columns, keeping the records per page to 25 or 50 and adding quick filters on the Lookup field.
8. Compare the time taken by the two scripts and the report loading time before and after the changes and note down the observations.

Script 1 - slow version (fetch everything and filter in the loop)

```
allOrders = Orders[ID != 0];
pendingTotal = 0;
for each ord in allOrders
{
    if(ord.Status == "Pending")
    {
        pendingTotal = pendingTotal + ord.Amount;
    }
}
info pendingTotal;
```

Script 2 - tuned version (criteria, range and aggregate function)

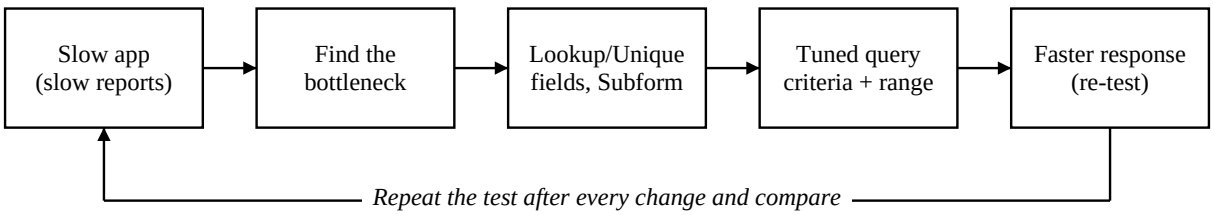
```
// use the criteria on the field so that only the needed records are fetched
pendingTotal = Orders[Status == "Pending"].sum(Amount);
pendingCount = Orders[Status == "Pending"].count();

// for a particular customer fetch only the first 50 records using range
custOrders = Orders[Customer == input.Customer] range from 1 to 50;
info pendingTotal;
info pendingCount;
info custOrders.count();
```

Note: a fetch inside a for each loop should be avoided because it runs once for every record. Fetch the records once before the loop, or use the aggregate functions (count, sum, max) directly. The time shown in the log changes with the load on the server, so the test can be repeated a few times.

DIAGRAM:

Fig 1: Performance optimization steps for a Zoho Creator application



RESULT:

Thus the performance optimization techniques were applied to a Zoho Creator application. The use of lookup and unique fields in the criteria, the subform for the item details and the tuned Deluge query with criteria, range and aggregate functions reduced the execution time of the script and the loading time of the report when compared with the original application.